AUTOSAR High-Level Design

EV Battery Monitoring and Thermal Management System

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Version: 1.0

Status: Approved for demonstration

Purpose: Synthetic training document for the AUTOSAR HLD Document Analysis Assistant project.

1. System Overview

The system monitors battery voltage, battery temperature and state of charge. The BatteryMonitor software component publishes battery status information. The ThermalController consumes battery temperature and generates a cooling request. The Dashboard consumes battery status for driver display.

2. High-Level Architecture

The application layer contains BatteryMonitor, ThermalController and Dashboard. Communication between application components is performed through defined ports and interfaces. The Runtime Environment (RTE) provides the communication mechanism between application software components.

3. Software Components

3.1 BatteryMonitor

BatteryMonitor acquires battery measurements and calculates BatteryStatus. It provides BatteryVoltage, BatteryTemperature and BatteryStatus information to other components.

3.2 ThermalController

ThermalController receives BatteryTemperature from BatteryMonitor and determines whether cooling is required. It publishes CoolingRequest to the cooling subsystem.

3.3 Dashboard

Dashboard receives BatteryStatus and presents battery information to the vehicle user.

4. Component Inventory

Component	Role	Inputs	Outputs
BatteryMonitor	Battery measurement and status	BatteryVoltage, BatteryTemperature	BatteryStatus
ThermalController	Thermal management	BatteryTemperature	CoolingRequest
Dashboard	Driver display	BatteryStatus	DisplayUpdate

5. Ports and Interfaces

5.1 BatteryMonitor Ports

P_BatteryVoltage provides battery voltage information. P_BatteryTemperature provides battery temperature information. P_BatteryStatus provides calculated battery status.

5.2 ThermalController Ports

R_BatteryTemperature receives battery temperature. P_CoolingRequest provides the requested cooling state.

5.3 Dashboard Ports

R_BatteryStatus receives BatteryStatus. P_DisplayUpdate provides the updated display information.

6. Interface Definitions

Interface	Data Elements	Provider	Consumer
BatteryMeasurementInterface	BatteryVoltage, BatteryTemperature	BatteryMonitor	ThermalController
BatteryStatusInterface	BatteryStatus	BatteryMonitor	Dashboard
CoolingRequestInterface	CoolingRequest	ThermalController	CoolingSystem

7. Signals and Data Flow

BatteryVoltage and BatteryTemperature are measurement data produced by BatteryMonitor. BatteryStatus is derived from battery measurements and is consumed by Dashboard. BatteryTemperature is consumed by ThermalController. CoolingRequest is generated by ThermalController.

8. Functional Flow

1. BatteryMonitor reads battery voltage and temperature. 2. BatteryMonitor calculates BatteryStatus. 3. BatteryMonitor provides BatteryTemperature to ThermalController. 4. ThermalController evaluates the temperature and generates CoolingRequest. 5. BatteryMonitor provides BatteryStatus to Dashboard. 6. Dashboard updates the driver display.

9. Dependency Mapping

ThermalController depends on BatteryMonitor for BatteryTemperature. Dashboard depends on BatteryMonitor for BatteryStatus. CoolingSystem depends on ThermalController for CoolingRequest.

10. Architecture Relationships

The primary application relationships are:

• BatteryMonitor → ThermalController through BatteryMeasurementInterface

• BatteryMonitor → Dashboard through BatteryStatusInterface

• ThermalController → CoolingSystem through CoolingRequestInterface

11. Validation Rules

Every consumed data element should have a defined provider. Every interface should identify its provider and consumer. Component names, port names and interface names should remain consistent across architecture sections.

12. Revision Information

Version 1.0 is the baseline demonstration revision. Future revisions may add new components, signals, interfaces or dependencies. The HLD analysis assistant should compare revisions and report potential additions, removals and inconsistencies for engineer review.

13. Example Questions for the AI Assistant

• Which component consumes BatteryTemperature?

• Which components depend on BatteryMonitor?

• What interface connects BatteryMonitor and Dashboard?

• What is the functional flow from battery measurement to cooling request?

• Which component provides BatteryStatus?